

19.4 THE DANGERS OF HIGH DEBT

Suppose that large deficits have led to a high debt ratio. What should the government do then? Simply trying to stabilize the debt at this high level is unwise: The lesson from history is that high debt can lead to vicious cycles and makes the conduct of fiscal policy extremely difficult. Let us look at this more closely.

Let us look at equation (19.16) again.

$$\frac{B_t}{Y_t} - \frac{B_{t-1}}{Y_{t-1}} = (r - g) \frac{B_{t-1}}{Y_{t-1}} + \frac{G_t - T_t}{Y_t} \quad \dots(19.16)$$

Let us take a country with a high debt ratio, say, 100%. Suppose the real interest rate is 3% and the growth rate is 2%. The first term on the right is (3%-2%) times 100% = 1% of GDP. Suppose further that the government is running a primary surplus of 1%, thus just enough to keep the debt ratio constant (the right side of the equation equals (3%-2%) times 100% + (-1%) = 0 per cent.

Now, suppose financial investors start to worry that the government may not be able to fully repay the debt. They ask for a higher interest rate, to compensate for what they perceive as a higher risk of default on the debt. But this in turn makes it more difficult for the government to stabilize the debt. Suppose, for example, that the interest rate increases from 3% to, say, 8%. Then, just to stabilize the debt, the government now needs to run a primary surplus of 6% (the right side of the equation is then equal to (8% - 2%) × 100% + (-6%) = 0 per cent.

Suppose that, in response to the increase in the interest rate, the government indeed takes measures to increase the primary surplus to 6 per cent. The spending cuts or tax increases that are needed are likely to prove politically costly, potentially generating more political uncertainty, a higher risk of default, and thus a further increase in the interest rate. Also, the sharp fiscal contraction is likely to lead to a recession, decreasing the growth rate. Both the increase in the real interest rate and the decrease in growth further increase (r - g), requiring an even larger surplus to stabilize the debt. At some point, the government may become unable to increase the primary surplus sufficiently, and the debt ratio starts increasing, leading financial markets to become even more worried and require an even higher interest rate. Increases in the interest rate and increases in the debt ratio feed on each other. The result is a debt explosion. The lesson is clear. When a government inherits a high debt ratio, it should aim at decreasing it over time. As (19.16) has made clear, it can achieve this through a combination of primary surpluses, high growth rates, and low real interest rates.

19.4.1 Debt Default

At some point, when a government finds itself facing very high interest rates, it may decide to default. Default is often partial, and creditors take what is known as a haircut: A haircut of 30 per cent, for example, means that creditors receive only 70 per cent of what they were owed. When debt is very high, default would seem like an appealing solution: Having a lower level of debt after default

reduces the size of the required fiscal consolidation and thus makes it more credible. It lowers required taxes, potentially allowing for higher growth. But default comes with very high costs. If debt is held, for example, by pension funds, the retirees may suffer very much from the default. If it is held by banks, then banks may go bankrupt, with major adverse effects on the economy. If debt is held instead mostly by foreigners, then the country's reputation may be lost, and it may be very difficult for the government to borrow abroad for a long time. So, in general, and rightly so, governments are very reluctant to default on their debt.

19.4.2 Lessons from the Past

High levels of government debt may increase the risk that an economy will experience capital flight that is an abrupt decline in the demand for a country's assets in world financial markets. International investors are aware that a government can always deal with its debt simply by defaulting. This approach was used as far back as 1335, when England's King Edward III defaulted on his debt to Italian bankers. More recently, several Latin American countries defaulted on their debts in the 1980s, and Russia did the same in 1998. The higher the level of the government debt, the greater is the temptation of default. Thus, as government debt increases, international investors may come to fear default and curtail their lending. If this loss of confidence occurs suddenly, the result could be the classic symptoms of 'capital flight' leading to currency crisis. It has occurred in Mexico in the early 1990s when default appeared likely. In 2011, it seemed likely that Greece was heading toward that outcome as well.

Greece's shaky financial status was well-known when it was admitted to the European Union (EU) in 2000, but the true depth of that nation's debt became known only in 2009 when Greece announced that it had been under-reporting its debt over the years. Meanwhile, fellow EU member Ireland, a strong financial performer, had to bail out its entire banking system in 2007-08. Between 2007 and 2010, the Irish debt-GDP ratio went up from 25 to about 100 percent, and its fiscal deficit as a percentage of GDP went up from zero per cent to more than 30 percent. At the same time, Portugal, Spain and Italy also started experiencing high debt problems. As the members of EU are connected, these nations' debt crisis threatened to trigger a sovereign debt crisis in Europe. Financial markets stopped viewing Italian, Greek, Irish, Portuguese and Spanish debt as close substitutes for German bonds, and they started hiking interest rates in the fall of 2008 to compensate for the heightened risk of default. By January 2012, the yield spread between German and Greek debt had increased by 33 percent. Greek banks then held about 20 percent of the government's sovereign debt. As the market feared Greece's banks would no longer honour obligations, they stopped rolling over Greek bank debt, which in turn meant that Greek banks could no longer roll over funding of the government's debt. A debt default seemed inescapable. To avert the crisis, the IMF, European Central Bank, and the European Commission agreed to extend emergency funding to Greece.

Since 2010, the EU and IMF have provided a total of more than €1 trillion in several loan packages to ease rollover problems, with Germany and France

contributing the most. The European Central Bank also injected up to €1 trillion of liquidity into the EU's banking system.

19.5 MONEY FINANCE

When the fiscal situation is bad, either because the deficits are large or the debt-GDP ratio is high, and the interest rate faced by the government is high, it becomes increasingly tempting for the government to want to finance itself through money finance. The government issues new bonds and instructs the central bank to buy them. The central bank then pays the government with the money it creates, and the government uses that money to finance its deficit. This process is called debt monetization.

How large a deficit can a government finance through such money creation? Let H be the amount of central bank money in the economy. Let ΔH be money creation; that is, the change in the nominal money stock from one month to the next. The revenue, in real terms that the government generates by creating an amount of money equal to ΔH is therefore $\Delta H/P$ – money creation during the period divided by the price level. This revenue from money creation is called seignorage.

$$\text{seignorage} = \frac{\Delta H}{P} \quad \dots(19.17)$$

To see what rate of (central bank) nominal money growth is required to generate a given amount of seignorage, we can rewrite $\Delta H/P$ as

$$\frac{\Delta H}{P} = \frac{\Delta H}{H} \frac{H}{P} \quad \dots(19.18)$$

From equation (19.8), we can think of seignorage ($\Delta H/P$) as the product of the rate of nominal money growth ($\Delta H/H$), and the real money stock (H/P). Thus we can re-write equation (19.17) as

$$\text{seignorage} = \frac{\Delta H}{H} \frac{H}{P} \quad \dots(19.19)$$

It is convenient to take one more step and divide both sides of the above equation by monthly GDP, say Y^I , to get:

$$\frac{\text{seignorage}}{Y^I} = \frac{\Delta H}{H} \left(\frac{H/P}{Y^I} \right) \quad \dots(19.20)$$

Suppose the government wants to finance a deficit of 10 per cent for a given ratio of central bank money to monthly GDP of 1. This implies that nominal money growth must satisfy

$$\frac{\Delta H}{H} \text{ times } 1 = 10\% \Rightarrow \frac{\Delta H}{H} = 10\%$$

From here we can infer that the monthly growth rate of nominal money of 10% is required to finance a deficit of 10 per cent. Unfortunately even such high rate of nominal money growth will not be sufficient to finance this deficit. As money growth increases, inflation is likely to follow. And very high inflation is likely to lead people to want to reduce their demand for money, and in turn the demand for central bank money. In other words, as the government increases ($\Delta H/H$), it is very likely that (H/P) will decrease. In response, the government needs to increase the rate of money growth further to achieve the same level of revenues. But higher money growth leads to further inflation, a further decrease in (H/P) and the need for further money growth. Soon, high inflation is likely to turn into

hyperinflation. You should note that several countries have witnessed hyperinflation in the past. Venezuela, for example, witnessed an inflation rate in the range of 3,00,000 per cent per year!

As inflation becomes very high, there is typically an increasing consensus that it should be stopped. Eventually, the government reduces the deficit and no longer has recourse to money finance. Inflation stops, but not before the economy has suffered substantial costs.

19.6 EFFECTS OF A DECREASE IN BUDGET DEFICIT

Suppose the government is running a budget deficit and decides to reduce it by decreasing its spending from G to G' while leaving taxes T unchanged. How will this affect the economy in the short run and in the medium run? Assume that output is initially at the natural level of output, so that the economy is at point A in Fig. 19.1. Output equals Y_n . The decrease in government spending from G to G' shifts the aggregate demand curve to the left, from AD to AD' . For a given price level, output is lower. In the short run, the equilibrium moves from A to A' ; output decreases from Y_n to Y' and the price level decreases from P to P' . The initial effect of the deficit reduction triggers lower output.

What happens over time? As long as output is below the natural level of output, we know that the aggregate supply curve keeps shifting down. The economy moves down along the aggregate demand curve AD' until the aggregate supply curve is given by AS'' and the economy reaches point A'' . By that time, the recession is over, and output is back at Y_n . At point A'' , not everything is the same as before: Output is back to the natural level of output, but the price level and the interest rate are lower than before the shift. The best way to see these specific effects is again to look at the adjustment in terms of the underlying $IS-LM$ model.

In panel (b) of Fig. 19.1, before the change in fiscal policy, the equilibrium is given by the intersection of the IS curve and the LM curve, at point A – which corresponds to point A in Fig. 19.1 panel (a). Output is equal to the natural level of output, Y_n , and the interest rate is given by i .

As the government reduces the budget deficit, the IS curve shifts to the left, to IS' . If the price level did not change, the economy would move from point A to point B . But, because the price level declines in response to the decrease in output, the real money stock increases, leading to a partially offsetting shift of the LM curve, down to LM' . So, the initial effect of deficit reduction is to move the economy from point A to point A' (Point A' in Fig. 19.1 panel (b) corresponds to point A' in Fig. 19.1 panel (a)).

Fig. 19.1 Panel (a): Decrease in Budget Deficit

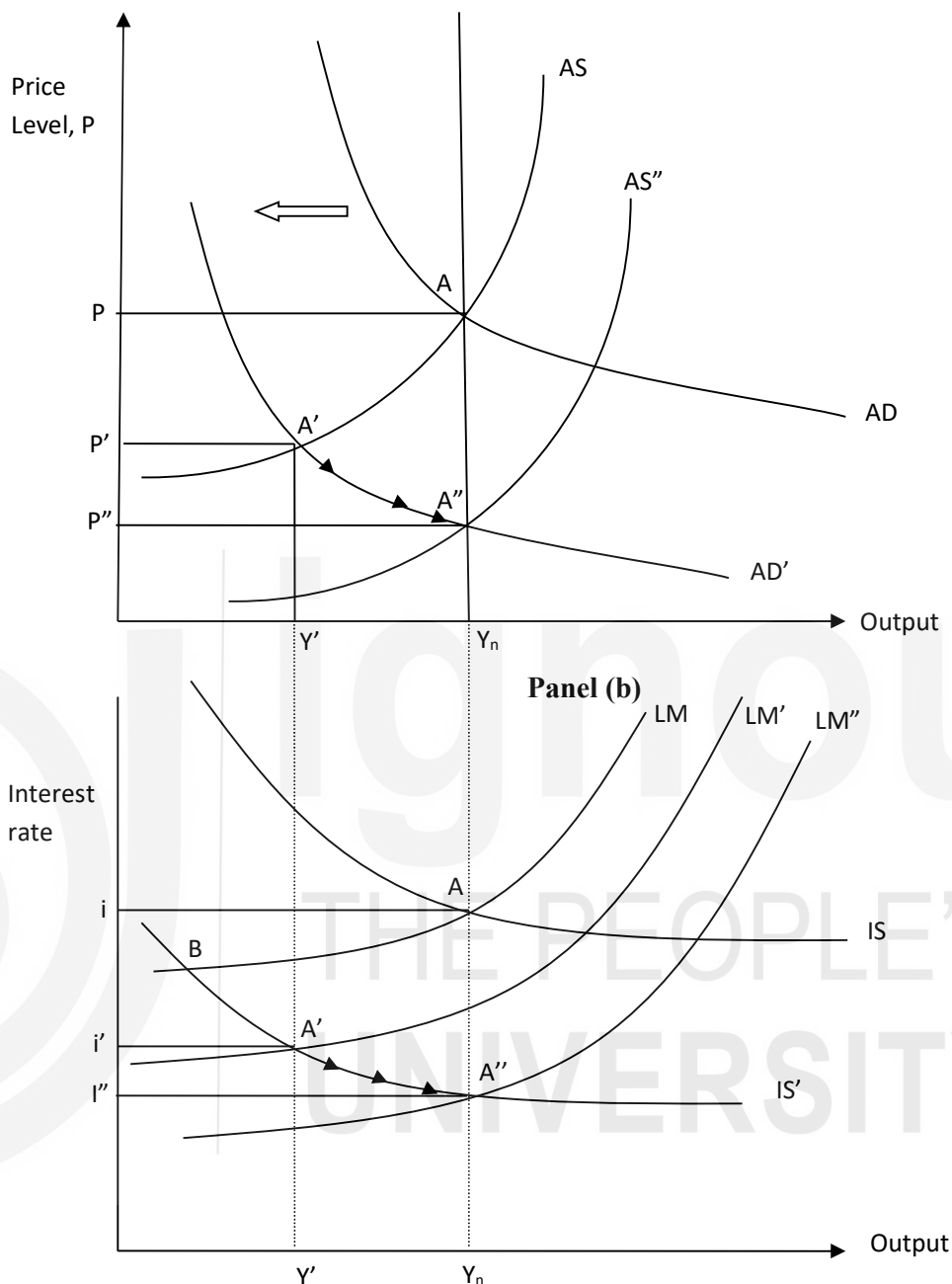


Fig. 19.1 shows that a deficit reduction leads in the short run to a decrease in output and to a decrease in the interest rate. In the medium run, output returns to its natural level, while the interest rate declines further.

Both output and the interest rate are lower than before the fiscal contraction. So long as output remains below the natural level of output, the price level continues to decline, leading to a further increase in the real money stock. The LM curve continues to shift down. In Fig. 19.1 panel (b), the economy moves down from point A' along IS' , and eventually reaches A'' (which corresponds to A'' in Fig. 19.1 panel (a)). At A'' , the LM curve is given by LM' . At A'' , output is back at the

natural level of output. But the interest rate is lower than it was before deficit reduction, down from i to i'' . Hence, in the medium run, a reduction in the budget deficit unambiguously leads to a decrease in the interest rate and an increase in investment.

Check Your Progress 3

- 1) Will a high rate of nominal money growth be sufficient to finance the fiscal deficit through seignorage?

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19.7 LET US SUM UP

As the government carries out its fiscal policy through its budget, a closely related concept is of fiscal sustainability, which is loosely defined as the government's ability to meet its current and future obligations. It is the sustainability of government's finances. The issue of fiscal sustainability has taken on special importance in the aftermath of the global financial crisis. A sustainable policy is one where the ratio of debt held by the public to GDP (the debt-GDP ratio) is ultimately stable or declining. Fiscal sustainability can be achieved by putting consistent effort in reducing distortions and waste in a tax system and its administration, broadening the tax base, eliminating wasteful government expenditures, and providing public goods and services that achieve their goal in an efficient way.

19.10 ANSWERS/ HINTS TO CHECK YOUR PROGRESS

Check Your Progress 1

- 1) The government budget constraint links the change in government debt to the initial level of debt (which affects interest payments) and to current government spending and taxes.

Check Your Progress 2

- 1) The increase in the ratio of debt to GDP will be larger: The higher the real interest rate, the lower the growth rate of output, the higher the initial debt ratio, the higher the ratio of the primary deficit to GDP.
- 2) The government must eliminate the deficit by running a primary surplus equal to the interest payments on the existing debt.

Check Your Progress 3

- 1) As money growth increases, inflation is likely to follow. And very high inflation is likely to lead people to want to reduce their demand for money, and in turn the demand for central bank money. In other words, as the government increases, $\Delta H/H$, H/P is likely to decrease. As it does so, the government needs to increase the rate of money growth further to achieve the same level of revenues. But higher money growth leads to further inflation, a further decrease in H/P and the need for further money growth.

